Renovation of Yamaha CA-800 II

George Biro

April 27, 2026

Abstract

1 Intoduction

The purchase price was 230 EUR plus shipping. Initially, the unit was operational, but the sound was not completely clean, and the feet were missing. All switches and potis were cleaned using Tesanol 26026. The power supply lamp was replaced, as it was broken.

2 Measurement and Test Equipment

Table 1: Measurement and Test Equipment Used

Instrument Type	**Model**
Signal Generator	UNI-T UTG932E
AD Converter	E1DA COSMOS ADC ISO Grade B (ES9822Pro, 32-bit, 768 kHz)
LCR Meter	DER EE DE-5000
Oscilloscope	Hameg HM303-4
Multimeter	UNI-T UT980C
Line-Phono Converter	OMNITRONIC LH-042
VAOhm Meter	Philips PM2503

3 Power Supply

Random spike noises were observed on the power supply output. The voltages at the emitters of TR701 and TR704 were measured at +50 V and -50 V respectively. Zener diodes D703, D704, D705, and D706 had previously been replaced by a third party, changing the original 31 V and 21 V types to 30 V and 20 V types. In general, this modification does not appear to have introduced any issues. Low hFE was observed in TR701 and TR704, and these transistors were replaced with BD139 and BD140 devices. An additional cooling

board was installed, and mechanical reinforcement of the main board was implemented to support it. Capacitors C705–C708 were replaced with Panasonic EEU-FS2A221 units, as the original capacitors exhibited significantly reduced capacitance (approximately 140 µF). Capacitors C709 and C710 were replaced with Panasonic EEUFR1J101L units. Transistors TR705, TR707, and TR709 (2SC458) were replaced with KSC1845F devices. Following these modifications, complete stability of the power supply was achieved.

4 Function Board

The function board contains the phono preamplifier and input selection circuitry. Some hum was present on this board. The capacitance and ESR of all electrolytic capacitors were measured, and any capacitor was replaced if a new component provided significantly lower ESR.

Modifications were required because the replacement transistors exhibit different internal capacitances. Circuit simulations indicated significant deviations when these transistors were used with the *Technics EPC-207C* moving-magnet cartridge. In addition, the original configuration resulted in unsatisfactory sound quality.

Given that the EPC-207C cartridge has an inductance of approximately 400 mH and an internal resistance of approximately 400 Ω, the modified RIAA resistor–capacitor network exhibits very low phase and gain errors. As a result, the overall sound quality was significantly improved. The modification was also simulated, and the results are shown in fig. 2.

Documentation Note At positions R409, R410, R427, and R428, the installed resistors were verified to be original components. Measured resistance values of 4.7 kΩ (R409 and R410) and 330 Ω (R427 and R428) were confirmed by both direct measurement and resistor color coding. The values specified in the official schematic differ from those observed and do not appear to be electrically consistent with correct circuit operation. A documentation error in the schematic is therefore suspected.

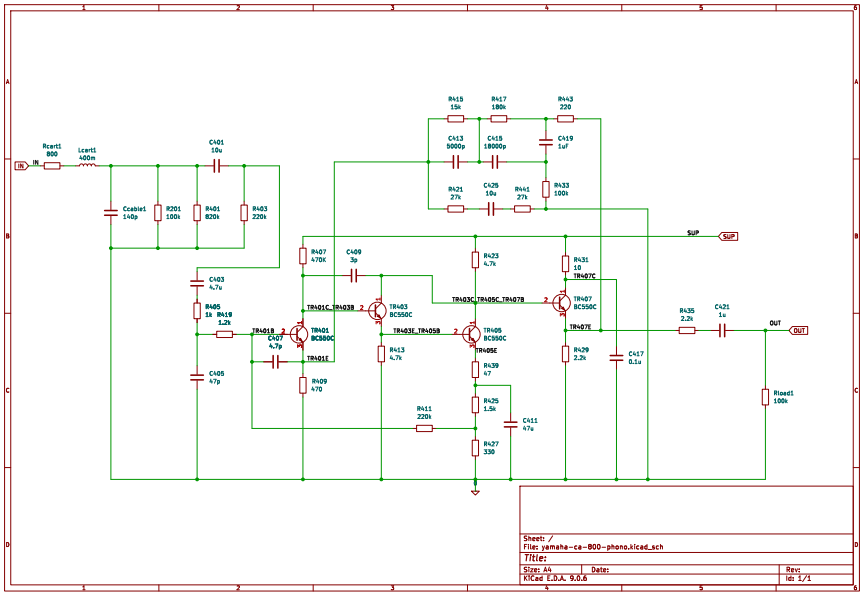

Figure 1: Schematics for simulation

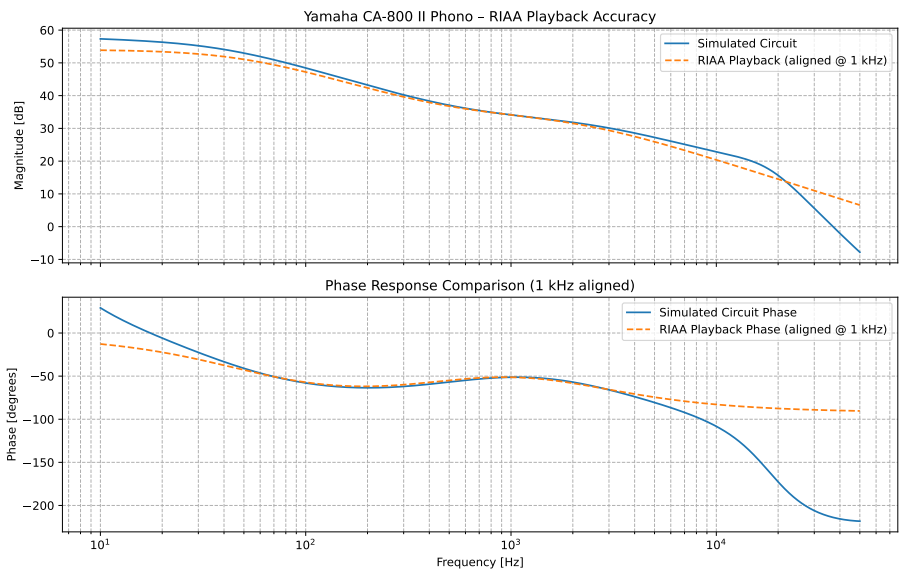

Figure 2: Simulation of the phono preamplifier with a 400 mH cartridge using the modified component values.

Table 2: Function Board Repair Log

Ref.	**Original**	**Replacement**
TR401, TR402	2SC1345	BC550C (HFE $\approx$ 600)
TR403, TR404	2SC1345	KSC1845 (HFE $\approx$ 400)
TR405, TR406	2SC1345	BC550C (HFE $\approx$ 600)
TR407, TR408	2SC1166	BC546C (HFE $\approx$ 600)
C405, C406	100 pF	3 pF
C407, C408	3 pF	47 pF
C409, C410	3 pF	47 pF
C411, C412		47 μF / 25 V (Kemet A750EK476M1EAAE040)
C401, C402		10 μF / 25 V (Kemet A758BG106M1EAAE070)
C403, C404		10 μF / 25 V (Kemet A758BG106M1EAAE070)
C413, C414	5000 pF	3300 pF
C415, C416	18 nF	12 nF
C417, C418	4.7 μF	1 μF / 50 V (Kemet ESC105M050AC3AA)
C241		220 μF / 50 V (Panasonic EEU-FR1H221B)
R413, R414	15 kΩ	18 kΩ
R415, R416	180 kΩ	330 kΩ

5 Tone control board

The capacitance and ESR of all electrolytic capacitors were measured, and any capacitor was replaced if a new component provided significantly lower ESR.

Table 3: Tone Control Board Repair Log

Ref.	**Original**	**Replacement**
TR301, TR302	2SC1345D	KSC1845F (HFE $\approx$ 400)
TR305..TR308	2SC1345D	KSC1845F (HFE $\approx$ 400)
C347		220 μF / 50 V (Panasonic EEU-FR1H221B)
C325, C325		10 μF / 25 V (Kemet A758BG106M1EAAE070)

6 Filter Board

The capacitance and ESR of all electrolytic capacitors were measured, and any capacitor was replaced if a new component provided significantly lower ESR.

Table 4: Filter Board Repair Log

Ref.	**Original**	**Replacement**
TR501, TR502	2SC1345D	KSC1845F (HFE $\approx$ 400)
TR505, TR506	2SC1345D	KSC1845F (HFE $\approx$ 400)
C526		220 μF / 50 V (Panasonic EEU-FR1H221B)
C325, C325		10 μF / 25 V (Kemet A758BG106M1EAAE070)

7 Final Measurements

7.1 Phono Stage

The signal generator was set to an output level of $450\,\mathrm{mV_{rms}}$ and connected to the input of the inverse RIAA converter, the output of which was connected to the Phono 1 input of the amplifier. An $8\,\Omega$ load resistor was connected to the amplifier output, and the volume control was adjusted to correspond to an output power of approximately 45 W. The same input voltage and volume setting were applied at all test frequencies.

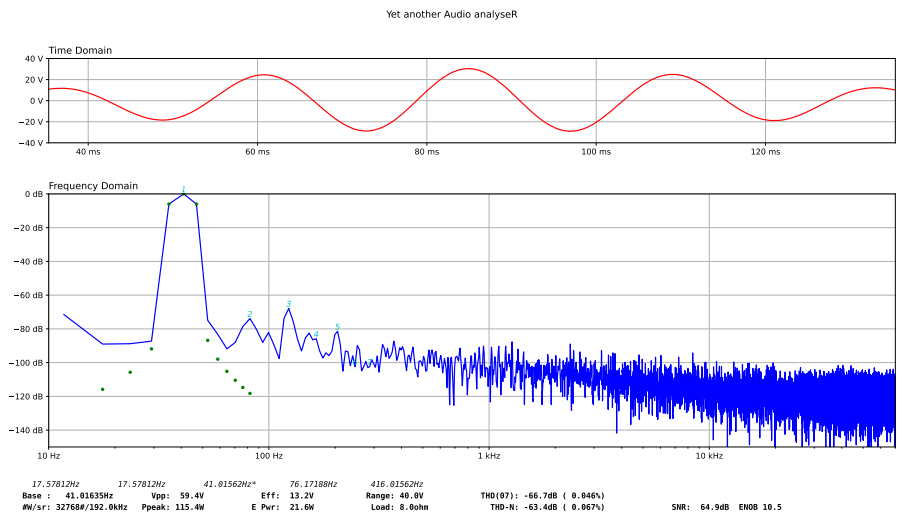

Figure 3: Phono 41Hz Left

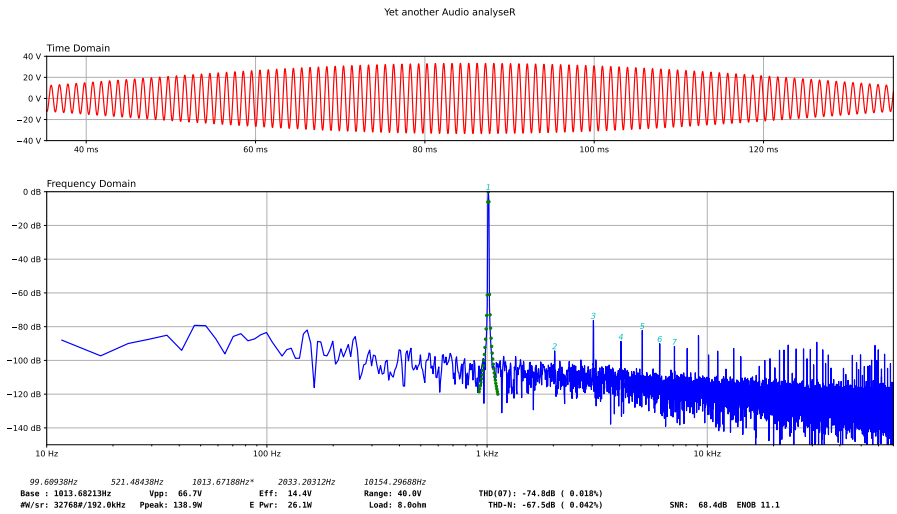

Figure 4: Phono 1kHz Left

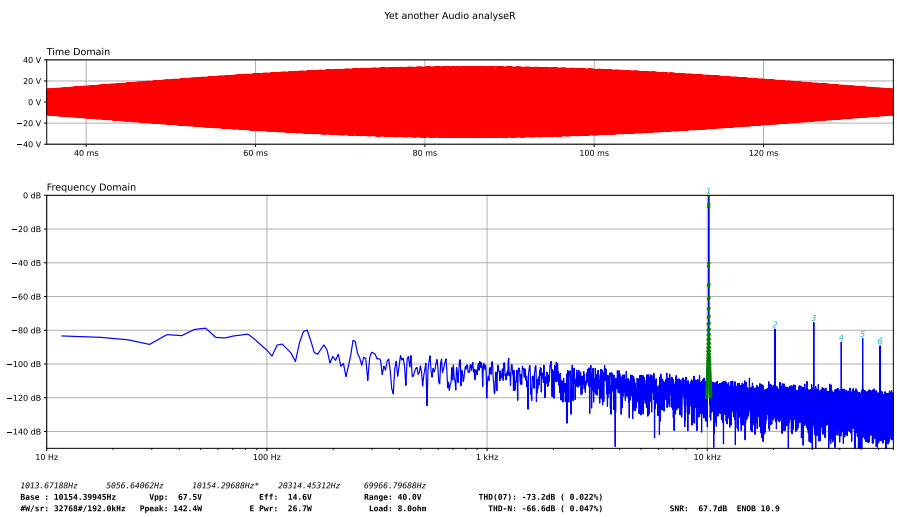

Figure 5: Phono 10kHz Left

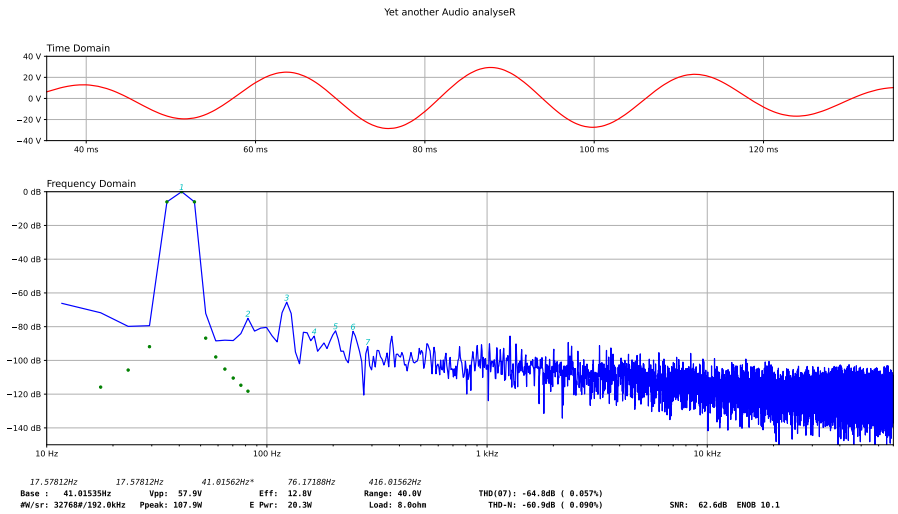

Figure 6: Phono 41Hz Right

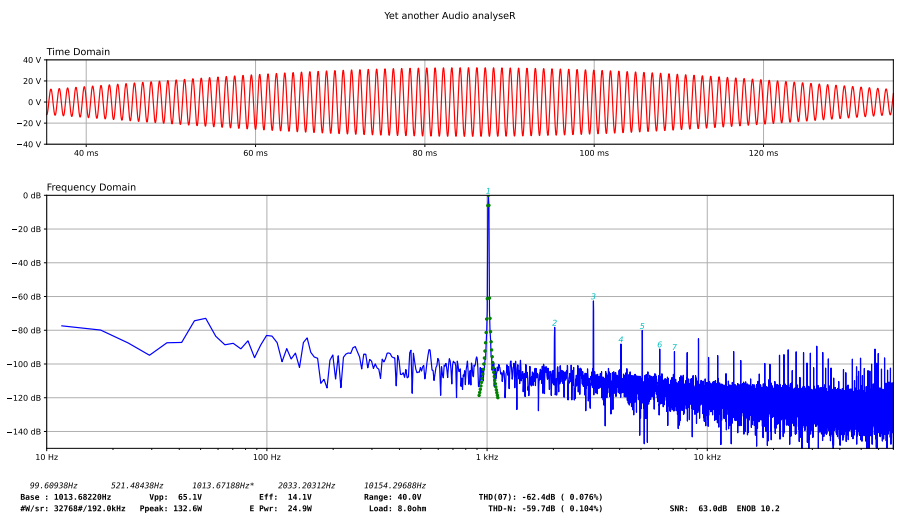

Figure 7: Phono 1kHz Right

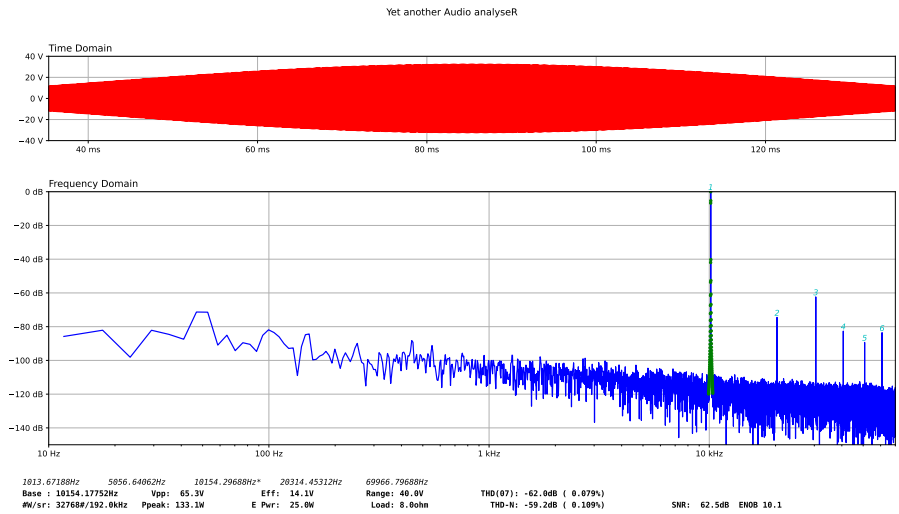

Figure 8: Phono 10kHz Right

7.2 Aux

The signal generator was set to an output level of $250\,\mathrm{mV_{rms}}$ and connected to the AUX1 input of the amplifier. An $8\,\Omega$ load resistor was connected to the amplifier output, and the volume control was adjusted to correspond to an output power of approximately 45 W. The same input voltage and volume setting were applied at all test frequencies.

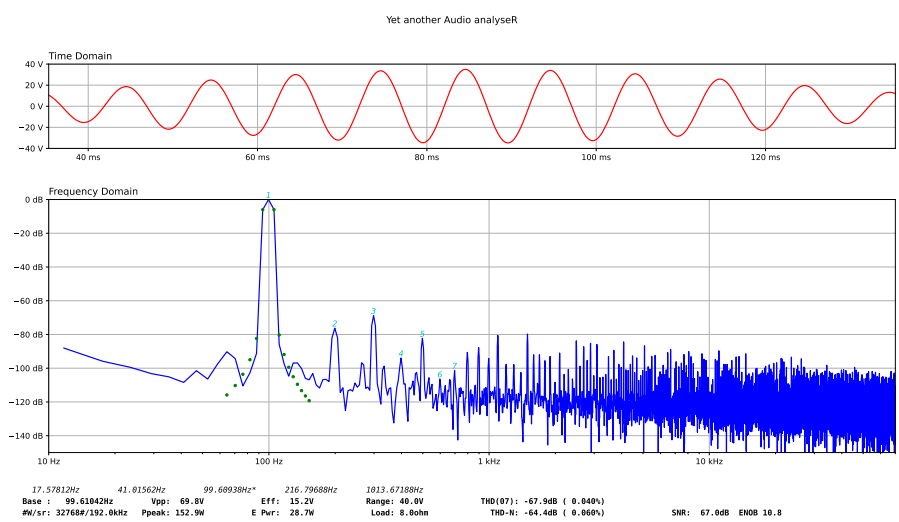

Figure 9: Aux 100Hz Left

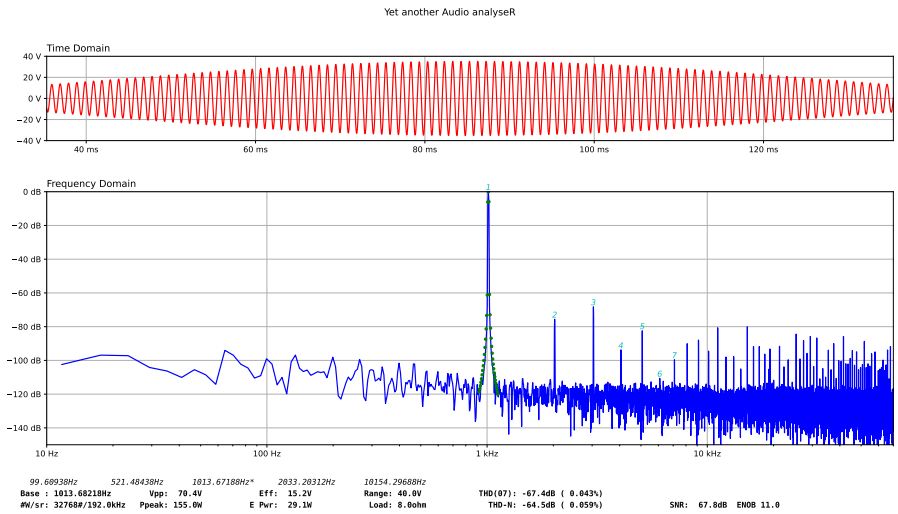

Figure 10: Aux 1kHz Left

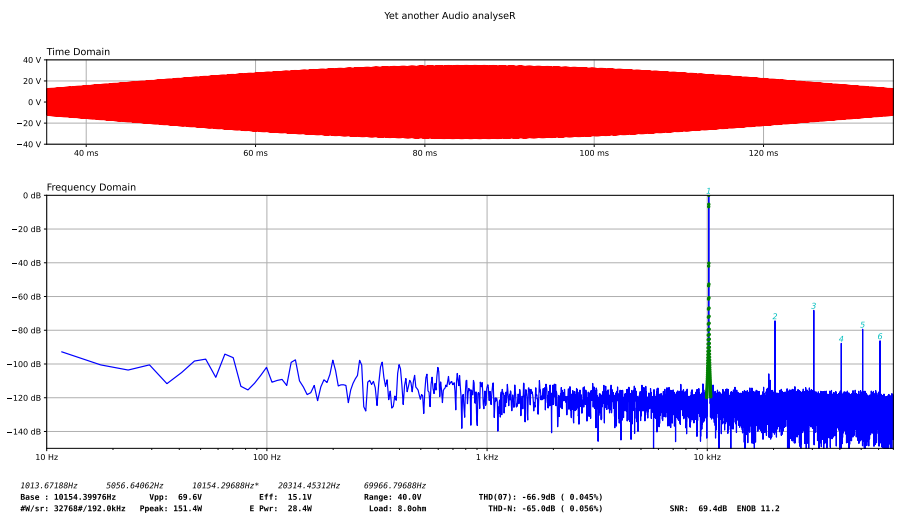

Figure 11: Aux 10kHz Left

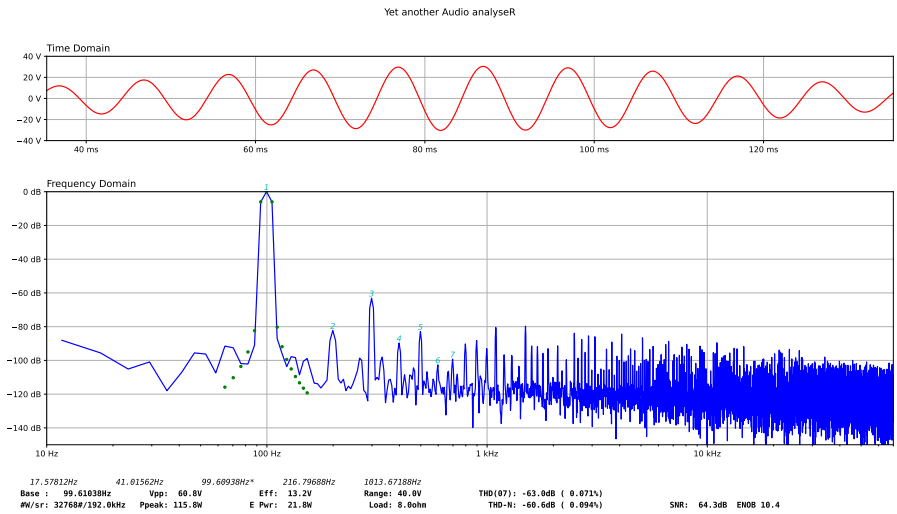

Figure 12: Aux 100Hz Right

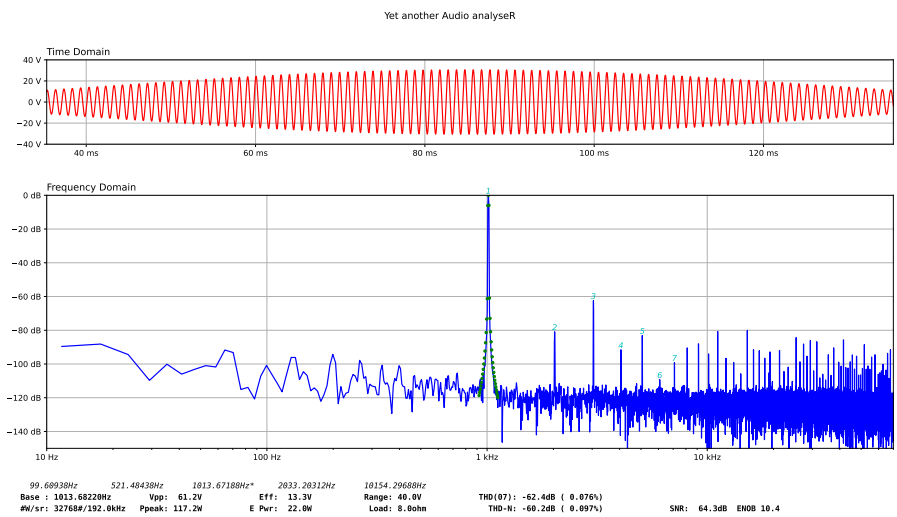

Figure 13: Aux 1kHz Right

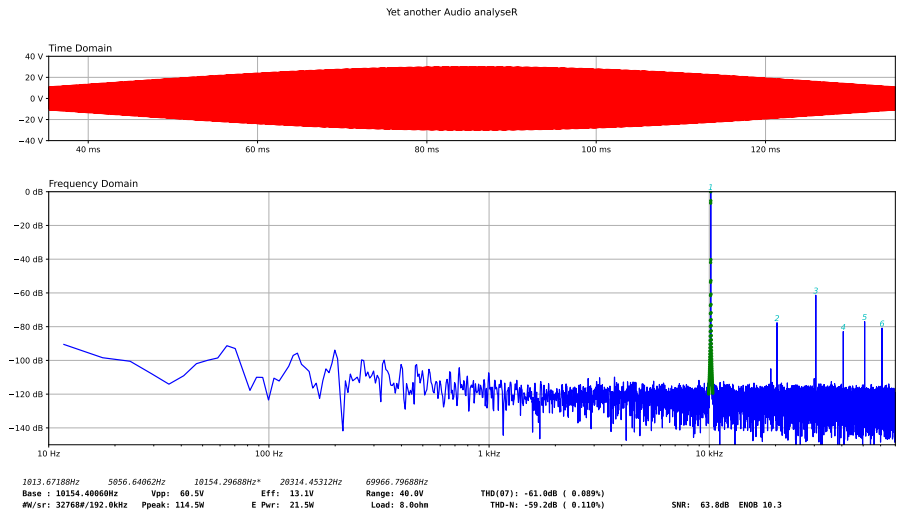

Figure 14: Aux 10kHz Right

8 Appendix I: Phone NGSpice simulation

```
.title KiCad schematic
* Official onsemi/Fairchild BC549/550, adjusted for BC550C-like gain
.param C405val = 47p
.param C407val = 3p
.param C409val = 3p

.model BC550C NPN (
+ LEVEL=1
+ IS=2.24183E-14 NF=0.996496 ISE=1.90217E-14 NE=2
+ BF=600 IKF=0.211766 VAF=161.939
+ NR=0.993 ISC=4.7863E-15 NC=0.996
+ BR=12.1807 IKR=0.3423 VAR=123.229
+ RB=167.033 IRB=7.079458E-05 RBM=1.12256
+ RE=0.036 RC=0.79
+ XTB=1.65 EG=1.1737 XTI=3
+ CJE=1.87E-11 VJE=0.732 MJE=0.33
+ CJC=6.16E-12 VJC=0.395 MJC=0.251 XCJC=0.6192 FC=0.5
+ TF=518.15E-12 XTF=10 VTF=10 ITF=1 TR=10.000E-9 )

.model KSC1845F npn (
+ IS=5.075431E-13 BF=400 NF=1
+ BR=0.365 NR=1 ISE=1.68107E-12 NE=2.0
+ ISC=1.8378E-12 NC=1.5
+ VAF=82.803 VAR=100 IKF=0.20596 IKR=0.0190546
```

```
+ RB=107 RBM=0.292 IRB=1.258925E-6
+ RE=0.15 RC=0.28
+ CJE=1.10447E-11 VJE=0.7300286 MJE=0.1619943 FC=0.5
+ CJC=5.625739E-12 VJC=0.45 MJC=0.3059045
+ XTB=1.7281 EG=1.1809 XTI=3
+ TF=1.108e-9 XTF=2.250 ITF=0.0044 VTF=4.0 TR=1.0e-8 )

R439 /TR405E Net-_C411-Pad1_ 47
R429 /TR407E 0 2.2k
R427 Net-_R411-Pad2_ 0 330
C411 Net-_C411-Pad1_ 0 47u
R425 Net-_C411-Pad1_ Net-_R411-Pad2_ 1.5k
R411 /TR401B Net-_R411-Pad2_ 220k
QTR401 /TR401C_TR403B /TR401B /TR401E BC550C
R409 /TR401E 0 470
R413 /TR403E_TR405B 0 4.7k
R407 SUP /TR401C_TR403B 470k
QTR403 /TR403C_TR405C_TR407B /TR401C_TR403B /TR403E_TR405B KSC1845F
R201 Net-_C401-Pad1_ 0 100k
R401 Net-_C401-Pad1_ 0 820k
R403 Net-_C401-Pad2_ 0 220k
C401 Net-_C401-Pad1_ Net-_C401-Pad2_ 10u
R443 Net-_C419-Pad1_ /TR407E 220
C419 Net-_C419-Pad1_ Net-_C415-Pad2_ 1u
R441 Net-_C425-Pad2_ 0 27k
C425 Net-_C425-Pad1_ Net-_C425-Pad2_ 10u
QTR407 /TR407C /TR403C_TR405C_TR407B /TR407E BC550C
R433 Net-_C415-Pad2_ 0 100k
R431 SUP /TR407C 10
QTR405 /TR403C_TR405C_TR407B /TR403E_TR405B /TR405E BC550C
R423 SUP /TR403C_TR405C_TR407B 4.7k
R421 /TR401E Net-_C425-Pad1_ 27k
C403 Net-_C403-Pad1_ Net-_C401-Pad2_ 4.7u
R405 Net-_C403-Pad1_ Net-_C405-Pad2_ 1k
R419 Net-_C405-Pad2_ /TR401B 1.2k
C417 /TR407C 0 0.1u
R435 /TR407E Net-_C421-Pad1_ 2.2k
C421 Net-_C421-Pad1_ OUT 1u
Rload1 OUT 0 100k

C407 /TR401B /TR401E {C407val}
C409 /TR401C_TR403B /TR403C_TR405C_TR407B {C409val}
C405 0 Net-_C405-Pad2_ {C405val}

Ccable1 Net-_C401-Pad1_ 0 140p
*Gardo
```

```
*Rcart1 IN Net-_Lcart1-Pad1_ 470
*Lcart1 Net-_Lcart1-Pad1_ Net-_C401-Pad1_ 60m
*Technics
Rcart1 IN Net-_Lcart1-Pad1_ 400
Lcart1 Net-_Lcart1-Pad1_ Net-_C401-Pad1_ 400m

* HF damping branch (tune these)
*Rhd  Net-_C401-Pad1_ n_hd 22k
*Chd  n_hd 0 220p


*C413 /TR401E Net-_C413-Pad2_ 5000p
*R415 /TR401E Net-_C413-Pad2_ 15k
*C415 Net-_C413-Pad2_ Net-_C415-Pad2_ 18000p
*R417 Net-_C413-Pad2_ Net-_C419-Pad1_ 180k

C413 /TR401E Net-_C413-Pad2_ 3300p
R415 /TR401E Net-_C413-Pad2_ 18k
C415 Net-_C413-Pad2_ Net-_C415-Pad2_ 12000p
R417 Net-_C413-Pad2_ Net-_C419-Pad1_ 330k

V1 SUP 0 51
V2 IN 0 AC 0.004
RLoad OUT 0 47k
Rleak_IN IN 0 100Meg
Rleak_OUT OUT 0 100Meg

.control
set filetype=ascii
op
print v(/TR401B) v(/TR401C_TR403B) v(/TR401E) v(/TR403C_TR405C_TR407B) v(/TR403E_TR405B) v(

* ==========================================================
* AC analysis
* ==========================================================
ac dec 300 10 50k

* ==========================================================
* Export raw data for Python
* ==========================================================
wrdata sim_ac.csv frequency v(out) v(in)
quit
.endc

.end
```

9 Appendix II: RIAA comparsion

```
import numpy as np
import matplotlib.pyplot as plt
import subprocess
import argparse
import sys

class CustomHelpFormatter(argparse.HelpFormatter):
def _get_help_string(self, action):
help = action.help
if action.default is not argparse.SUPPRESS:
help += f' (default: {action.default})'
return help

# Begin of the program
parser = argparse.ArgumentParser(
prog='Yet Another Audio Analyzer',
usage='%(prog)s [options]',
formatter_class=CustomHelpFormatter)

parser.add_argument("--psfile", type=str, default="pspice-yamaha-ca800-phono.cir")
parser.add_argument("--acfile", type=str, default="sim_ac.csv")
parser.add_argument(
"--param",
action="append",
default=[],
help="ngspice parameter definition (passed as --define=...)"
)
args = parser.parse_args()
cmd = [
"ngspice",
"-b"]

for p in args.param:
cmd.append(f"--define={p}")
cmd.append(args.psfile)
print(cmd)

subprocess.run(cmd, check=True)

# ============================================================
# Load ngspice wrdata CSV (Mac format: all vectors are complex)
#
# Columns:
# 0 : freq (real)
```

```
# 1 : Re(freq)   [ignore]
# 2 : Im(freq)   [ignore]
# 3 : freq (vout) [ignore]
# 4 : Re(Vout)
# 5 : Im(Vout)
# 6 : freq (vin)  [ignore]
# 7 : Re(Vin)
# 8 : Im(Vin)
# ============================================================
data = np.loadtxt(args.acfile)

freq = data[:, 0]                        # Hz
Vout = data[:, 4] + 1j * data[:, 5]
Vin  = data[:, 7] + 1j * data[:, 8]

# H = Vin / Vout
H = Vout / Vin

mag_db = 20 * np.log10(np.abs(H))
phase_deg = np.unwrap(np.angle(H)) * 180 / np.pi


# ============================================================
# Exact RIAA PLAYBACK (de-emphasis) transfer function
# ============================================================

# RIAA time constants:
# 3180 µs → 50.05 Hz
# 318  µs → 500.5 Hz
# 75   µs → 2122 Hz

f1 = 50.05     # Hz
f2 = 500.5     # Hz
f3 = 2122.0    # Hz

w = 2 * np.pi * freq
w1 = 2 * np.pi * f1
w2 = 2 * np.pi * f2
w3 = 2 * np.pi * f3

# Playback (DE-emphasis)
H_riaa = (1 + 1j*w/w2) / ((1 + 1j*w/w1) * (1 + 1j*w/w3))

mag_db_riaa = 20 * np.log10(np.abs(H_riaa))
phase_deg_riaa = np.unwrap(np.angle(H_riaa)) * 180 / np.pi
```

```
# ============================================================
# Normalize MAGNITUDE and PHASE at 1 kHz (RIAA reference)
# ============================================================

idx_1k = np.argmin(np.abs(freq - 1000))

# Magnitude normalization
mag_offset_db = mag_db[idx_1k] - mag_db_riaa[idx_1k]
mag_db_riaa_offset = mag_db_riaa + mag_offset_db

# Phase normalization
phase_offset = phase_deg[idx_1k] - phase_deg_riaa[idx_1k]
phase_deg_riaa_offset = phase_deg_riaa + phase_offset

mag_err = mag_db - mag_db_riaa_offset
phase_err = phase_deg - phase_deg_riaa_offset

print("Errors")
for f in [100, 500, 1000, 2000, 5000, 10000, 20000]:
idx_f = np.argmin(np.abs(freq - f))
print(f"{mag_err[idx_f]:8.2f}dB {phase_err[idx_f]:8.2f}deg @ {f:8.0f}Hz")


# ============================================================
# Combined magnitude + phase figure
# ============================================================

fig, (ax_mag, ax_phase) = plt.subplots(
2, 1, figsize=(11, 7), sharex=True
)

# ----------------------------
# Magnitude response subplot
# ----------------------------
ax_mag.semilogx(freq, mag_db, label="Simulated Circuit")
ax_mag.semilogx(
freq, mag_db_riaa_offset, "--",
label="RIAA Playback (aligned @ 1 kHz)"
)
ax_mag.set_ylabel("Magnitude [dB]")
ax_mag.set_title("Yamaha CA-800 II Phono { RIAA Playback Accuracy")
ax_mag.grid(True, which="both", ls="--")
ax_mag.legend()

# ----------------------------
# Phase response subplot
```

```
# ----------------------------
ax_phase.semilogx(freq, phase_deg, label="Simulated Circuit Phase")
ax_phase.semilogx(
freq, phase_deg_riaa_offset, "--",
label="RIAA Playback Phase (aligned @ 1 kHz)"
)
ax_phase.set_xlabel("Frequency [Hz]")
ax_phase.set_ylabel("Phase [degrees]")
ax_phase.set_title("Phase Response Comparison (1 kHz aligned)")
ax_phase.grid(True, which="both", ls="--")
ax_phase.legend()

plt.tight_layout()
plt.savefig("riaa.pdf")
plt.show()
```